Continuity of the t-Norm Is Necessary in Two Broad t-Conorm Regimes

Math discovery candidate partial result

August 17, 2026

1 Status and source question

Lai and Luo prove that if a t-norm T is continuous and a right-continuous t-conorm L satisfies their strictness condition (LS), then the tensor product $L \otimes T$ preserves the continuous distance distribution functions $\mathcal{D}_c^+$. Remark 4.7 asks whether continuity of T is necessary [1]. The question appears on printed page 15:

□

Theorem 4.6. *Let T be a continuous t-norm on $[0,1]$ and L be a right continuous t-conorm on $[0,\infty]$. Then $L \otimes T(f,g) \in \mathcal{D}_c^+$ for all $f, g \in \mathcal{D}_c^+$ if and only if L satisfies the condition (LS). In this case, $(\mathcal{D}_c^+, L\otimes T)$ is a subsemigroup, but not a submonoid of $(\Delta^+, L \otimes T)$ since the identity $\varepsilon(0,1)$ is not in $\mathcal{D}_c^+$.*

Proof. The "only if" part is from Proposition 4.3. To show the "if" part, since the condition (LS) is satisfied, by Theorem 4.5, $L\otimes T(f,g)$ is already in $\mathcal{D}_0^+$. Additionally, it always holds that $L\otimes T(f,g) \le f$. Hence, $L\otimes T(f,g)(0^+) = f(0^+) = 0$, which means that $L \otimes T(f,g)$ is continuous at 0. Therefore, $L \otimes T(f,g)$ is also continuous if both f and g are. □

Remark 4.7. In the above theorem, we assume that the t-norm T is continuous, but whether it is necessary remains open.

Corollary 4.8. *Let T be a continuous t-norm on $[0,1]$ and L be a continuous t-conorm on $[0,\infty]$. Then $L \otimes T(f,g) \in \mathcal{D}_{sc}^+$ for all $f, g \in \mathcal{D}_{sc}^+$ if and only if L satisfies the condition (LS) and T satisfies the condition*

$$(TS) \qquad x < y, x' < y' \Longrightarrow T(x,y) < T(x',y').$$

This packet proves necessity exactly for the maximum t-conorm. It also excludes every left-continuous t-norm having a nilpotent-minimum ordinal-sum component when L is addition, or more generally has an additive generator. The arbitrary-L question remains open.

2 Definitions and proof intuition

A distance distribution function (d.d.f.) is an increasing, left-continuous map $f : [0,\infty] \to [0,1]$ with $f(0) = 0$ and $f(\infty) = 1$. Write $\mathcal{D}_c^+$ for the continuous d.d.f.s. For finite x,

$$(L \otimes T)(f,g)(x) = \sup_{L(r,s)<x} T(f(r), g(s)). \tag{1}$$

For $L = M^*$, where $M^*(r,s) = \max\{r,s\}$, the source observes that (1) collapses to the pointwise composition $T(f(x), g(x))$. Continuous d.d.f.s can pass through any prescribed point of the unit square, so preservation of continuity tests all northeast limits of T. Left continuity already controls southwest limits; monotonicity then squeezes every other approach.

For addition, a nilpotent-minimum component has a threshold: below it the t-norm stays at the component floor, but immediately above it the value jumps to the midpoint. A piecewise-linear d.d.f. turns that algebraic threshold into a literal spatial discontinuity of (1).

3 The exact maximum-t-conorm subcase

Theorem 1 (Maximum t-conorm)**.** *Let T be a left-continuous t-norm. Then*

$$(\mathcal{D}_c^+, M^* \otimes T) \text{ is closed under } M^* \otimes T \quad \iff \quad T \text{ is continuous on } [0,1]^2.$$

Proof. The source paper computes, for every finite x,

$$(M^* \otimes T)(f,g)(x) = \sup_{r,s<x} T(f(r), g(s)) = T(f(x), g(x)). \tag{2}$$

If T is continuous, (2) is continuous on the finite half-line for every $f, g \in \mathcal{D}_c^+$. At infinity, $f(x), g(x) \to 1$, so the limit is $T(1,1) = 1$; hence the output lies in $\mathcal{D}_c^+$.

Conversely, assume the stated closure. Fix $p, q \in (0,1)$, and define continuous d.d.f.s

$$f_p(x) = \begin{cases} px, & 0 \le x \le 1, \\ p + (1-p)(x-1), & 1 \le x \le 2, \\ 1, & x \ge 2, \end{cases} \qquad g_q(x) = \begin{cases} qx, & 0 \le x \le 1, \\ q + (1-q)(x-1), & 1 \le x \le 2, \\ 1, & x \ge 2. \end{cases} \tag{3}$$

By closure and (2), $x \mapsto T(f_p(x), g_q(x))$ is continuous at 1. Therefore

$$\lim_{t \downarrow 0} T\big(p + (1-p)t, q + (1-q)t\big) = T(p,q). \tag{4}$$

Because T is increasing in each coordinate, any path approaching (p,q) strictly from the northeast is squeezed between two reparameterizations of the path in (4). Thus (4) is the full northeast limit of T at (p,q). The assumed left continuity gives

$$\lim_{t \downarrow 0} T(p-t, q-t) = T(p,q), \tag{5}$$

with truncated coordinates near the boundary.

Now let $(u_n, v_n) \to (p,q)$. For every sufficiently small $\delta > 0$, eventually

$$T(p-\delta, q-\delta) \le T(u_n, v_n) \le T(p+\delta, q+\delta), \tag{6}$$

again truncating at 0 and 1. Equations (4)–(5), followed by $\delta \downarrow 0$, squeeze $T(u_n, v_n)$ to $T(p,q)$. If $p = 0$ or $q = 0$, continuity follows directly from $0 \le T(u,v) \le \min\{u,v\}$. If $p = 1$ or $q = 1$, use the left-continuity squeeze in that coordinate and $T(1,v) = v$ in the other. Thus T is continuous everywhere. ☐

4 An explicit ordinal-sum obstruction

Let the nilpotent-minimum t-norm be

$$N(u,v) = \begin{cases} 0, & u+v \le 1, \\ \min\{u,v\}, & u+v > 1. \end{cases} \tag{7}$$

We say that a t-norm T has a nilpotent-minimum ordinal-sum component on $[a,b]$, $0 \le a < b \le 1$, if on that square it is

$$T(u,v) = a + (b-a) N\left(\frac{u-a}{b-a}, \frac{v-a}{b-a}\right), \tag{8}$$

and the usual ordinal-sum rule applies outside a common component (so the value is the minimum).

Theorem 2 (Addition)**.** *If a left-continuous t-norm T has a nilpotent-minimum ordinal-sum component, then $\mathcal{D}_c^+$ is not closed under $+ \otimes T$.*

Proof. For the component $[a, b]$, define the continuous d.d.f.

$$F(x) = \begin{cases} ax, & 0 \le x \le 1, \\ a + (b-a)(x-1), & 1 \le x \le 2, \\ b + (1-b)(x-2), & 2 \le x \le 3, \\ 1, & x \ge 3. \end{cases} \tag{9}$$

Set $H = (+ \otimes T)(F, F)$. We claim $H(3) = a$. Indeed, if $r + s < 3$ and both $r, s \in [1, 2]$, their normalized coordinates in (8) sum to less than 1, so $T(F(r), F(s)) = a$. If one spatial coordinate is below 1, one t-norm input is at most a. If one is above 2, the other must be below 1, and the ordinal-sum rule again gives a value at most a. Hence the supremum in (1) is at most a, while pairs in $[1, 2]^2$ with sum below 3 show that it is exactly a.

For every $x > 3$, choose $\varepsilon > 0$ with $2(3/2 + \varepsilon) < x$. Both spatial coordinates then lie in the component, their normalized sum is greater than 1, and

$$H(x) \ge T\big(F(3/2 + \varepsilon), F(3/2 + \varepsilon)\big) = a + (b-a)(1/2 + \varepsilon). \tag{10}$$

Consequently $\liminf_{x \downarrow 3} H(x) \ge a + (b-a)/2 > a = H(3)$. Thus $H \notin \mathcal{D}_c^+$. □

For $a = 0, b = 1$, (9) can be shortened to $F(x) = \min\{x, 1\}$; then $(+ \otimes N)(F, F)(1) = 0$, while its right limit is at least 1/2.

Corollary 3 (Additive-generator t-conorms)**.** *Suppose a t-conorm on the extended half-line has the form*

$$L(r, s) = \phi^{-1}\big(\phi(r) + \phi(s)\big), \tag{11}$$

where $\phi : [0, \infty] \to [0, \infty]$ is an increasing homeomorphism. If T has a nilpotent-minimum ordinal-sum component, then $\mathcal{D}_c^+$ is not closed under $L \otimes T$.

Proof. Let F be (9) and put $f = F \circ \phi$. This is a continuous d.d.f. For $X < \infty$, the inequality $L(r, s) < \phi^{-1}(X)$ is equivalent to $\phi(r) + \phi(s) < X$. Therefore

$$(L \otimes T)(f, f)(\phi^{-1}(X)) = (+ \otimes T)(F, F)(X).$$

The jump at $X = 3$ from Theorem 2 is transported to $\phi^{-1}(3)$. □

5 What remains open

Theorem 1 completely answers Remark 4.7 for one noncancellative t-conorm satisfying (LS). Theorem 2 and Corollary 3 rule out a canonical, broad discontinuous family for all cancellative t-conorms with additive generators. They do not cover every discontinuous left-continuous t-norm or every right-continuous t-conorm satisfying (LS).

The source's quasi-inverse formula suggests the missing general step. A discontinuity changes a strict superlevel set of T, but one would need to prove that this change is detected by

$$\inf_{T(p,q) > y} L(u(p), v(q)) \tag{12}$$

for some strictly increasing, infimum-preserving u, v. For a general L, other points of the superlevel set may keep the infimum in (12) unchanged. Eight focused upgrade attempts did not remove this order-separation obstruction.

6 Verification and novelty status

The maximum proof was audited at interior and boundary points. The ordinal witness was checked over all spatial cases $r + s < 3$, and the generator corollary is an exact change of variables. The accompanying verification report records those checks.

Bounded searches through August 17, 2026 by arXiv id, exact title, exact remark wording, and the relevant t-norm/t-conorm phrases found the final journal publication but no later answer to Remark 4.7. The source already contains identity (2); the exact continuity equivalence and the explicit ordinal-sum obstruction were not located. This is therefore a candidate new partial result, not a certified novelty claim.

References

[1] H. Lai and Q. Luo, *Triangle functions generated by tensor products of quantales*, arXiv:2411.11876 (2024); Fuzzy Sets and Systems 508 (2026), 109655.

[2] E. P. Klement, R. Mesiar, and E. Pap, *Triangular Norms*, Kluwer Academic Publishers, 2000.

[3] B. Schweizer and A. Sklar, *Probabilistic Metric Spaces*, Dover, 2005.